Deliquescence Humidity Test-containers

(Please observe the respective hazard warnings!)

The sensor accuracy can be checked using deliquescence humidity containers. These containers hold a saturated salt solution, over whose surface an equilibrium humidity is established. Various relatively harmless household salts and other substances, such as water and desiccants, can be used:

H_2O (dihydrogen monoxide, destilated water) results in 100% relative humidity.

NaCl (sodium chloride, table salt) results in 76% relative humidity.

K_2CO_3 (potassium carbonate, potash) results in 43% relative humidity (slight deviation in testing).

$MgCl_2$ (magnesium chloride, bath salts) results in 34% relative humidity.

LiCl (lithium chloride, desiccant) results in 11% relative humidity (not tested due to high cost).

$CaCl_2$ (dried calcium chloride, desiccant) results in 0–3% relative humidity.

All substances used are readily available and either already present in the household (water, table salt) or can be ordered, for example, from Amazon.